

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027****Vision:** Dream of where you want.**Mission:** Means to achieve Vision**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)**Keywords of POs:**

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	2026-27 (ODD)	Course Name	Lab : Embeded Systems
Semester	7	Course Code	23IOT1706
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	02
Course Outcome	CO1: Explain the fundamentals of embedded microcontrollers, development environments, and simulation tools. CO2: Develop embedded applications using Arduino/ESP32 and interface sensors, actuators, and peripherals. CO3: Implement event-driven and communication-based embedded systems using interrupts, UART, and I2C protocols. CO4: Design, simulate, and test embedded system applications for real-world applications.
Aim	To design and implement an interrupt-driven embedded application using an Arduino/ ESP32 (or ARM Cortex-M) that immediately responds to an emergency switch press by activating an alarm and sending an alert message.
Theory (100 words)	An interrupt is a mechanism that allows a microcontroller to respond immediately to an external or internal event without continuously checking for it. During normal operation, the microcontroller executes the main program, but when an interrupt occurs, it temporarily pauses the current task and executes a special function called an Interrupt Service Routine (ISR). After completing the ISR, the microcontroller resumes the main program. In an emergency alert system, a push button or emergency switch can be connected to an interrupt-capable digital pin of an Arduino or ESP32. When the switch is pressed, an interrupt is generated and the ISR detects the emergency condition. The controller can then activate an LED, buzzer, or other alarm device. It can also send an alert to a monitoring system using communication technologies such as Wi-Fi. Interrupt-based systems provide fast response, efficient CPU utilization, and reliable event handling, making them suitable for security systems, medical alarms, industrial safety systems, and emergency assistance applications.



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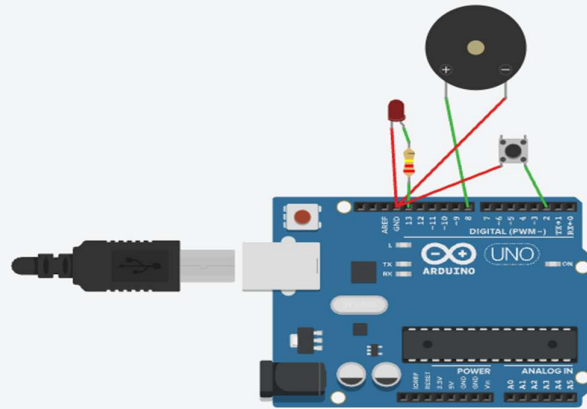
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Procedure and Execution (100 Words)	1. Connect the Arduino/ESP32 to the computer using a USB cable. 2. Connect an emergency push button to an interrupt-capable digital pin of the microcontroller. 3. Connect an LED and buzzer to suitable digital output pins. 4. Open the Arduino IDE and select the appropriate board and communication port. 5. Define the emergency switch pin as an interrupt input using the required interrupt function. 6. Write an Interrupt Service Routine (ISR) to detect the emergency button press. 7. Program the system to activate the LED and buzzer when an emergency interrupt occurs. 8. Add Serial Monitor or Wi-Fi alert functionality if an emergency notification is required. 9. Compile the program and check for any errors. 10.Upload the program to the Arduino/ESP32. 11.Run the system and press the emergency switch. 12.Observe whether the LED and buzzer are activated immediately. 13.Check the Serial Monitor or connected application for the emergency alert message. 14.Test the system multiple times to verify that the interrupt is detected quickly and reliably. 15.Record the output, observations, and results in the practical record.
	Stepwise Screenshots with steps:

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Magnificent Wolt

Circuits Schematic Components

Code Start Simulation Send To

1 (Arduino Uno R3)

```
1 //sheikh wasimuddin ES 2 61
2 volatile bool emergency = false;
3
4 void emergencyISR() {
5   emergency = true;
6 }
7
8 void setup() {
9   pinMode(2, INPUT_PULLUP); // Emergency button
10  pinMode(13, OUTPUT); // LED
11  pinMode(8, OUTPUT); // Buzzer
12
13  attachInterrupt(digitalPinToInterrupt(2), emergencyISR, FALLING)
14
15  Serial.begin(9600);
16 }
17
18 void loop() {
19   if (emergency) {
20     digitalWrite(13, HIGH); // Turn ON LED
21     digitalWrite(8, HIGH); // Turn ON buzzer
22
23     Serial.println("EMERGENCY ALERT!");
24
25     delay(3000);
26
27     digitalWrite(13, LOW); // Turn OFF LED
28     digitalWrite(8, LOW); // Turn OFF buzzer
29
30     emergency = false;
31   }
32 }
```



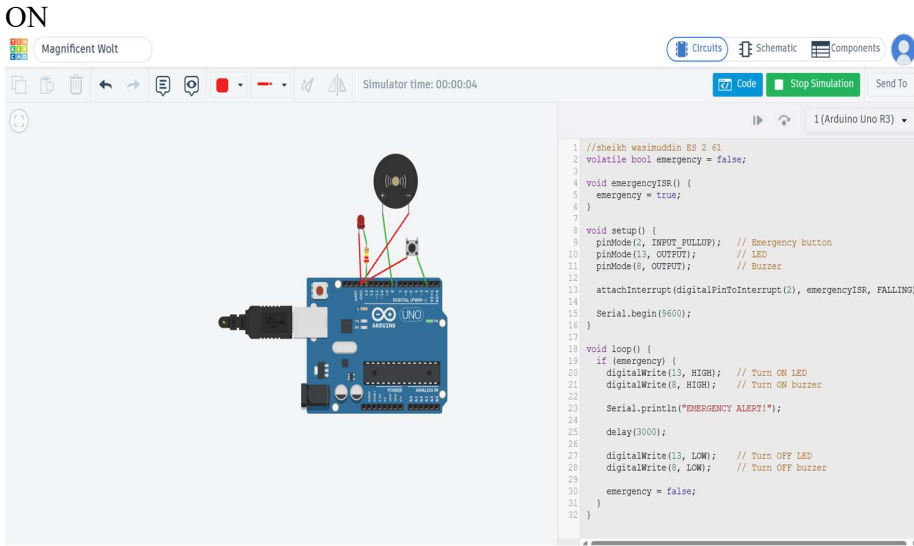
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Conclusion	The interrupt-based emergency alert system was successfully implemented using a microcontroller. An emergency push button was connected as an interrupt input, while an LED and buzzer were used to indicate the emergency condition. When the button was pressed, the interrupt was triggered immediately and the system activated the alarm. The emergency status was also displayed through the Serial Monitor. This practical provided a clear understanding of Interrupt Service Routines (ISR), interrupt-driven programming, and their importance in developing fast and reliable real-time safety and emergency applications.
Date	20-08-2026